Experiment No:2
Upload and Analyze the data set given in csv format and performance data preprocessing and visualization.

AIM:

To Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

DESCRIPTION:

Use sample data set sales-data.csv.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the data into a pandas DataFrame
file_path = '/content/sales_data.csv'
df = pd.read_csv(file_path)

# Display the first few rows
print(df.head())

# Check for missing values
print(df.isnull().sum())

# Fill missing Sales values with the mean
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())

# Drop rows with missing Product, Quantity, or Region
```

```
df.dropna(subset=['Product', 'Quantity', 'Region'],
inplace=True)

# Summary statistics
print(df.describe())

# Group by Product and calculate total Sales and Quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print(product_summary)

# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'],
product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

# Convert Date column to datetime
df['Date'] = pd.to_datetime(df['Date'])

# Line plot of sales over time
sales_over_time = df.groupby('Date').agg({
    'Sales': 'sum'
}).reset_index()
```

```
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'],
sales_over_time['Sales'], marker='o')
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc=np.sum,
    fill_value=0
)

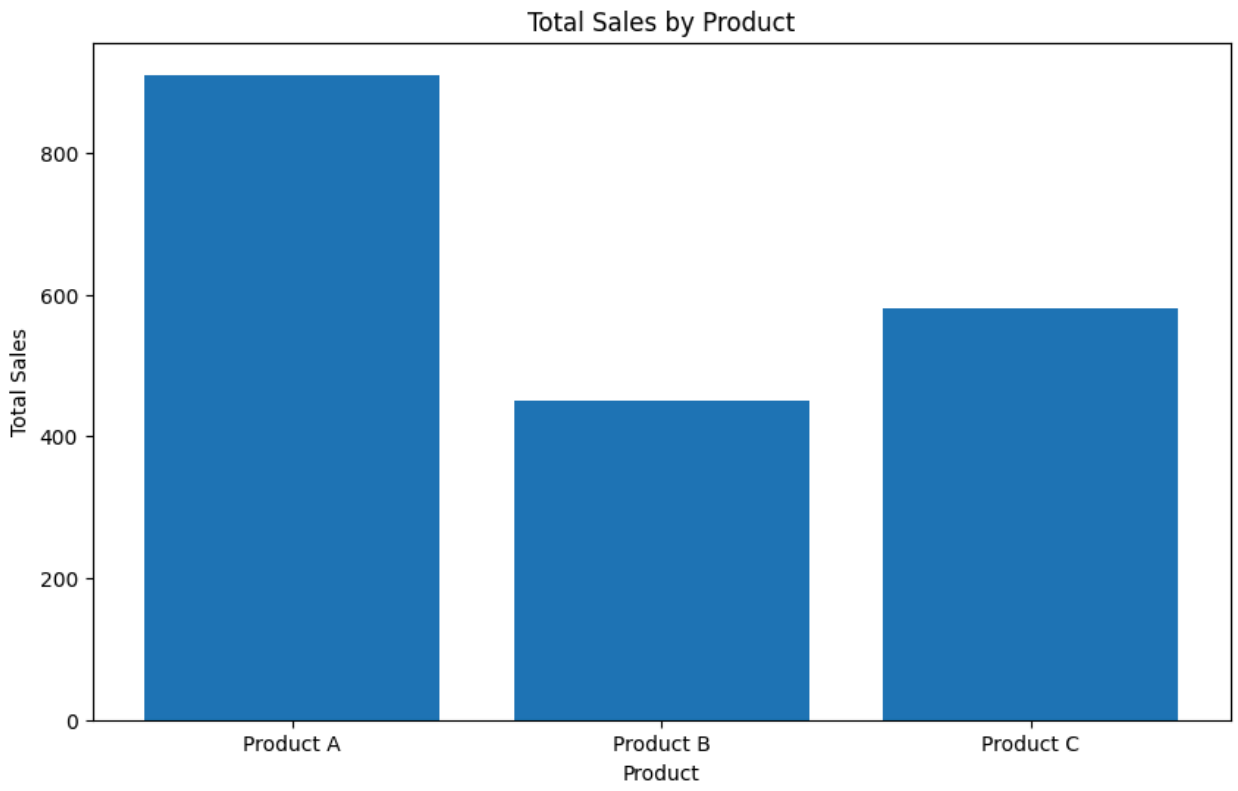
print(pivot_table)

# Correlation matrix for numeric columns only
correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)

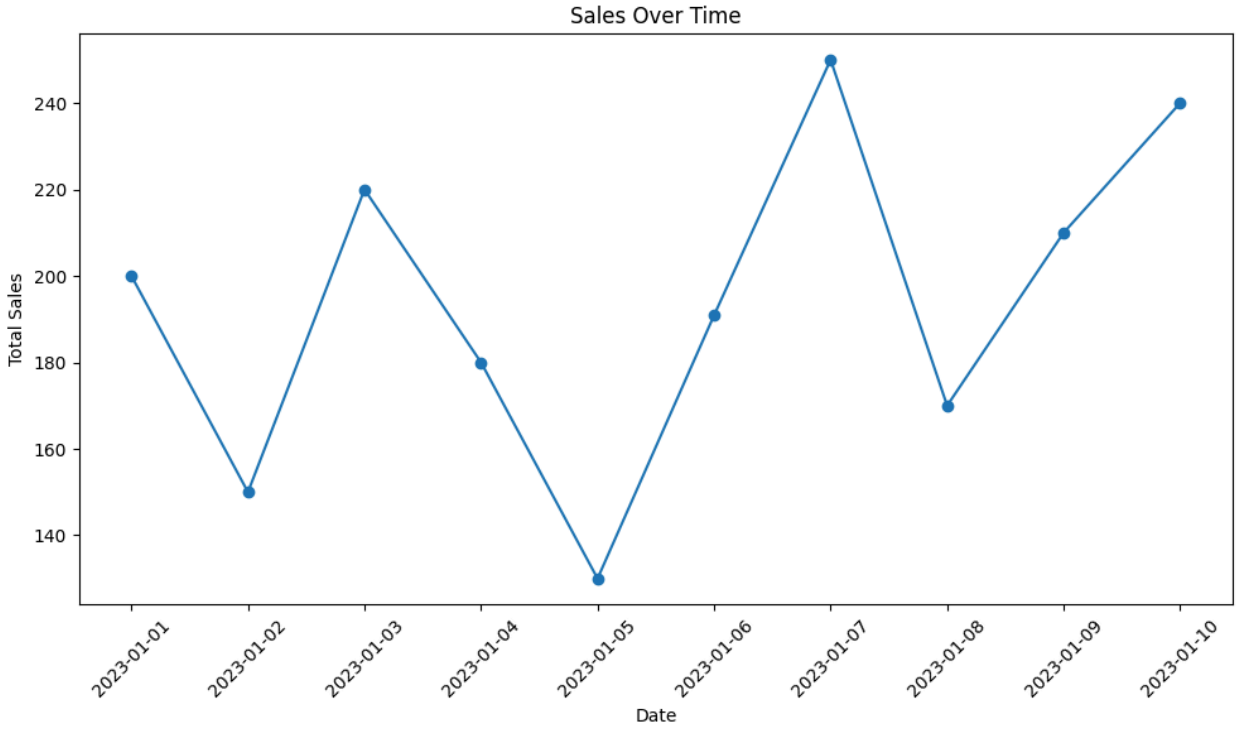
# Heatmap of the correlation matrix
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True,
cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

OUTPUT:

```
        Date    Product  Sales  Quantity Region
0  2023-01-01  Product A  200.0       4.0  North
1  2023-01-02  Product B  150.0       3.0  South
2  2023-01-03  Product A  220.0       5.0  North
3  2023-01-04  Product C  180.0       2.0   East
4  2023-01-05  Product B  130.0       2.0  South
Date        0
Product     0
Sales       1
Quantity    1
Region      1
dtype: int64
            Sales   Quantity
count   10.000000  10.000000
mean   194.090909   3.800000
std     38.054743   1.316561
min    130.000000   2.000000
25%    172.500000   3.000000
50%    195.454545   4.000000
75%    217.500000   4.750000
max    250.000000   6.000000
     Product       Sales  Quantity
0  Product A  910.000000      19.0
1  Product B  450.000000       8.0
2  Product C  580.909091      11.0
```




```
/tmp/ipykernel_1895/1755118830.py:59
```



```
: FutureWarning: The provided callable <function sum at
0x7f59fe584ea0> is currently using DataFrameGroupBy.sum. In a
future version of pandas, the provided callable will be used
directly. To keep current behavior pass the string "sum"
instead.
  pivot_table = df.pivot_table(
```

Product	Product A	Product B	Product C
Region			
East	0.0	0.0	580.909091
North	910.0	0.0	0.000000
South	0.0	450.0	0.000000

	Sales	Quantity
Sales	1.000000	0.838703
Quantity	0.838703	1.000000



RESULT:

Thus, the sales dataset was successfully uploaded and analyzed using Pandas. Data preprocessing was performed by handling missing values, and data visualization was carried out using a bar chart, line chart, and correlation heatmap. Product-wise sales and quantity totals were calculated, and a pivot table was created to analyze sales by region and product.